# *High prevalence and abundant atypical genotypes of Toxoplasma gondii isolated from lambs destined for human consumption in the USA.*

Little information is available on the presence of viable Toxoplasma gondii in tissues of lambs worldwide. The prevalence of T. gondii was determined in 383 lambs (<1 year old) from Maryland, Virginia and West Virginia, USA. Hearts of 383 lambs were obtained from a slaughter house on the day of killing. Blood removed from each heart was tested for antibodies to T. gondii by using the modified agglutination test (MAT). Sera were first screened using 1:25, 1:50, 1: 100 and 1:200 dilutions, and hearts were selected for bioassay for T. gondii. Antibodies (MAT, 1:25 or higher) to T. gondii were found in 104 (27.1%) of 383 lambs. Hearts of 68 seropositive lambs were used for isolation of viable T. gondii by bioassay in cats, mice or both. For bioassays in cats, the entire myocardium or 500g was chopped and fed to cats, one cat per heart and faeces of the recipient cats were examined for shedding of T. gondii oocysts. For bioassays in mice, 50g of the myocardium was digested in an acid pepsin solution and the digest inoculated into mice; the recipient mice were examined for T. gondii infection. In total, 53 isolates of T. gondii were obtained from 68 seropositive lambs. Genotyping of the 53 T. gondii isolates using 10 PCR-restriction fragment length polymorphism markers (SAG1, SAG2, SAG3, BTUB, GRA6, c22-8, c29-2, L358, PK1 and Apico) revealed 57 strains with 15 genotypes. Four lambs had infections with two T. gondii genotypes. Twenty-six (45.6%) strains belong to the clonal Type II lineage (these strains can be further divided into two groups based on alleles at locus Apico). Eight (15.7%) strains belong to the Type III lineage. The remaining 22 strains were divided into 11 atypical genotypes. These results indicate high parasite prevalence and high genetic diversity of T. gondii in lambs, which has important implications in public health. We believe this is the first in-depth genetic analysis of T. gondii isolates from sheep in the USA.